# Capstone Project –
# Online Retail Store Customer Segmentation

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# Problem Statement

The online retail industry collects a vast amount of transactional data every day. However, without properly analysing this data, the business cannot leverage it to understand customer behaviour, preferences, and value to the company. A major challenge faced by the retailer is the inability to differentiate between various types of customers such as high-value repeat buyers versus one-time shoppers. This limits the effectiveness of personalized marketing and retention strategies. Therefore, identifying customer segments is crucial to business growth.

# Project Objective

The objective of this project is to perform customer segmentation using clustering algorithms based on the purchasing patterns observed in the online retail dataset. This involves analysing customer behaviour across dimensions such as how recently they purchased (Recency), how often they purchase (Frequency), and how much they spend (Monetary value). Using these patterns, we aim to group customers into meaningful segments that can guide targeted marketing strategies and improve customer retention.

# Data Description

The dataset used in this project is the 'OnlineRetail.csv' file, which contains 387,961 rows and 8 columns. Each row corresponds to an item purchased in an invoice.

Each record includes the following features:-

| Feature Name | Description |
|---|---|
| InvoiceNo | A unique identifier for each transaction |
| StockCode | Product ID |
| Description | Name of the product |
| Quantity | Number of items purchased |
| InvoiceDate | Date and time of purchase |
| UnitPrice | Price per unit |
| CustomerID | Unique ID assigned to each customer |
| Country | Country of purchase |

The dataset contains transactions from a UK-based online retailer from December 2010 to December 2011.

*Table 1: Dataset Description*

| | Quantity | InvoiceDate | UnitPrice | CustomerID | TotalPrice |
|---|---|---|---|---|---|
| **count** | 397924 | 397924 | 397924 | 397924 | 397924 |
| **mean** | 13.02182 | 43:36.9 | 3.116174 | 15294.31517 | 22.394749 |
| **min** | 1 | 01-12-2010 08:26 | 0 | 12346 | 0 |
| **25%** | 2 | 07-04-2011 11:12 | 1.25 | 13969 | 4.68 |
| **50%** | 6 | 31-07-2011 14:39 | 1.95 | 15159 | 11.8 |
| **75%** | 12 | 20-10-2011 14:33 | 3.75 | 16795 | 19.8 |
| **max** | 80995 | 09-12-2011 12:50 | 8142.75 | 18287 | 168469.6 |
| **std** | 180.4202 | NaN | 22.09679 | 1713.169877 | 309.05559 |

# Data Preprocessing Steps And Inspiration

To ensure meaningful analysis, several preprocessing steps were applied:

1. Removed rows with missing Customer IDs to ensure customers can be tracked individually.
2. Filtered out cancelled transactions and those with non-positive quantities.
3. Created a 'TotalPrice' column as Quantity UnitPrice to represent transaction value.
4. Generated RFM metrics:
    - Recency: Days since last purchase
    - Frequency: Number of purchases
    - Monetary: Total amount spent
5. Scaled RFM values using Min-Max normalization for better performance in clustering.

These steps were inspired by best practices in customer analytics and RFM modeling.

# Choosing the Algorithm For the Project

K-Means clustering was chosen due to its effectiveness in grouping data based on numerical features such as RFM metrics. It is a widely-used, unsupervised machine learning algorithm that aims to partition customers into distinct groups based on similarity. With the ability to scale to large datasets and produce easily interpretable results, K-Means is ideal for the segmentation task.

K-Means clustering provides an efficient and intuitive way to segment customers based on purchasing behaviour.
Key reasons for its selection include:

- Speed and scalability for large retail datasets
- Easy to implement and interpret
- Well-suited for numerical features like Recency, Frequency, and Monetary
- Commonly used in marketing and CRM domains for customer profiling

It also allows us to easily visualize clusters in 2D or 3D space if needed.

# Assumptions

Several assumptions were made during the analysis:

- Each CustomerID represents a unique customer across all transactions.
- Transactions with missing or erroneous data were excluded to maintain data quality.
- Purchase behaviour over the 1-year period is representative of overall customer trends.
- Clustering results are based purely on transactional data, without demographic inputs.

# Model Evaluation and Technique

The effectiveness of the clustering model was evaluated using the Elbow Method, which involves plotting the within-cluster sum of squares (WCSS) against the number of clusters. The 'elbow point' on the curve indicates the optimal number of clusters. In this case, 4 clusters were selected.

After training the K-Means model, each customer was assigned to a cluster and cluster-wise RFM statistics were computed. These were used to name the clusters and derive actionable insights about each group.

# Inferences from the Project

Based on the K-Means clustering, the following customer segments were identified:

1. **Loyal Customers**: High frequency, high monetary, low recency; best for targeted campaigns and loyalty rewards.
2. **Churned Customers**: High monetary but low recency; likely lost customers that may respond to reactivation offers.
3. **Regular Shoppers**: Moderate in all dimensions; consistent buyers who can be nurtured for upselling.
4. **Low-Value or New Customers**: Low on all fronts; may need engagement or promotional incentives to grow their value.

These insights can help tailor marketing strategies, improve engagement, and boost customer lifetime value.

# Future Possibilities

1. Enhance segmentation using customer demographics, browsing behaviour, or product categories.
2. Integrate the model with a live dashboard for real-time segment tracking.
3. Use advanced clustering methods like DBSCAN or Gaussian Mixture Models for non-linear segments.
4. Apply A/B testing to assess the impact of segment-specific marketing campaigns.
5. Predict future behaviour using time series or deep learning models to anticipate churn or high-value customers.

# Conclusion

The customer segmentation project successfully leveraged the K-Means algorithm and RFM model to uncover meaningful patterns in customer behaviour. By dividing the customer base into distinct segments, the business gains a data-driven foundation for strategic decision-making. This segmentation approach helps focus marketing efforts, improve retention, and maximize customer lifetime value.

# References


- 'Data Science for Business' by Foster Provost and Tom Fawcett
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- RFM Customer Segmentation: https://towardsdatascience.com/
- UCI Repository  Online Retail Dataset